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Lec # 07Day: MTWTFSS

Exp:- find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ If z is defined

implicitly as a function of x and y by equation $x^3 + y^3 + z^3 + 6xyz + 4 = 0$, then evaluate these partial derivatives at the point $(-1, 1, 2)$.

Solution:-

$$\frac{\partial z}{\partial x} = - \frac{F_x}{F_z} = - \frac{\text{derivative of g.f w.r.t } x}{\text{derivative of g.f w.r.t } z}$$

$$\frac{\partial z}{\partial x} = - \frac{3x^2 + 0 + 0 + 6yz}{0 + 0 + 3z^2 + 6xy}$$

$$= - \frac{3x^2 + 6yz}{3z^2 + 6xy}$$

$$= - \frac{x^2 + 2yz}{z^2 + 2xy}$$

$$\frac{\partial z}{\partial x} = \left(\frac{x^2 + 2yz}{z^2 + 2xy} \right)$$

$$\frac{\partial z}{\partial y} = - \frac{F_y}{F_z}$$

$$= \frac{\partial z}{\partial y} = - \frac{0 + 3y^2 + 0 + 6xz + 0}{0 + 0 + 3z^2 + 6xy + 0}$$

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$$= \frac{-3y^2 + 6xz}{3x^2 + 6xy}$$

$$= \frac{-3(y^2 + 2xz)}{3(x^2 + 2xy)}$$

$$= \boxed{\frac{-y^2 + 2xz}{x^2 + 2xy}}$$

$\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(-1, 1, 2)$

$$\left. \frac{\partial z}{\partial x} \right|_{(-1, 1, 2)} = \frac{-(x^2 + 2yz)}{(x^2 + 2xy)}$$

$$= \frac{-((-1)^2 + 2(1)(2))}{(2)^2 + 2(-1)(1)} = \frac{-(1+4)}{4-2} = \boxed{-\frac{5}{2}}$$

$$\left. \frac{\partial z}{\partial y} \right|_{(-1, 1, 2)} = \frac{-(y^2 + 2xz)}{(x^2 + 2xy)}$$

$$= \frac{-(1)^2 + 2(-1)(2)}{(2)^2 + 2(-1)(2)}$$

$$= \frac{-1 - 4}{4 - 2} = \boxed{-\frac{5}{2}}$$